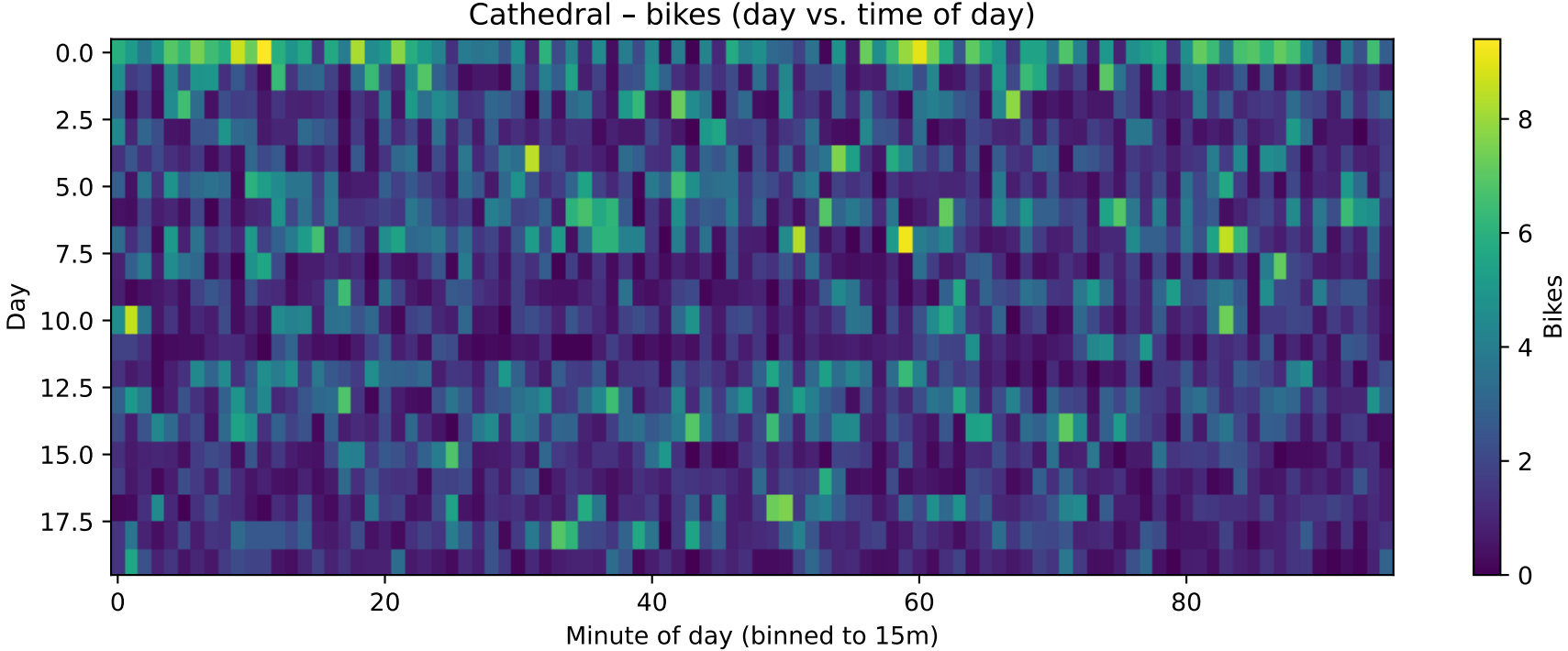
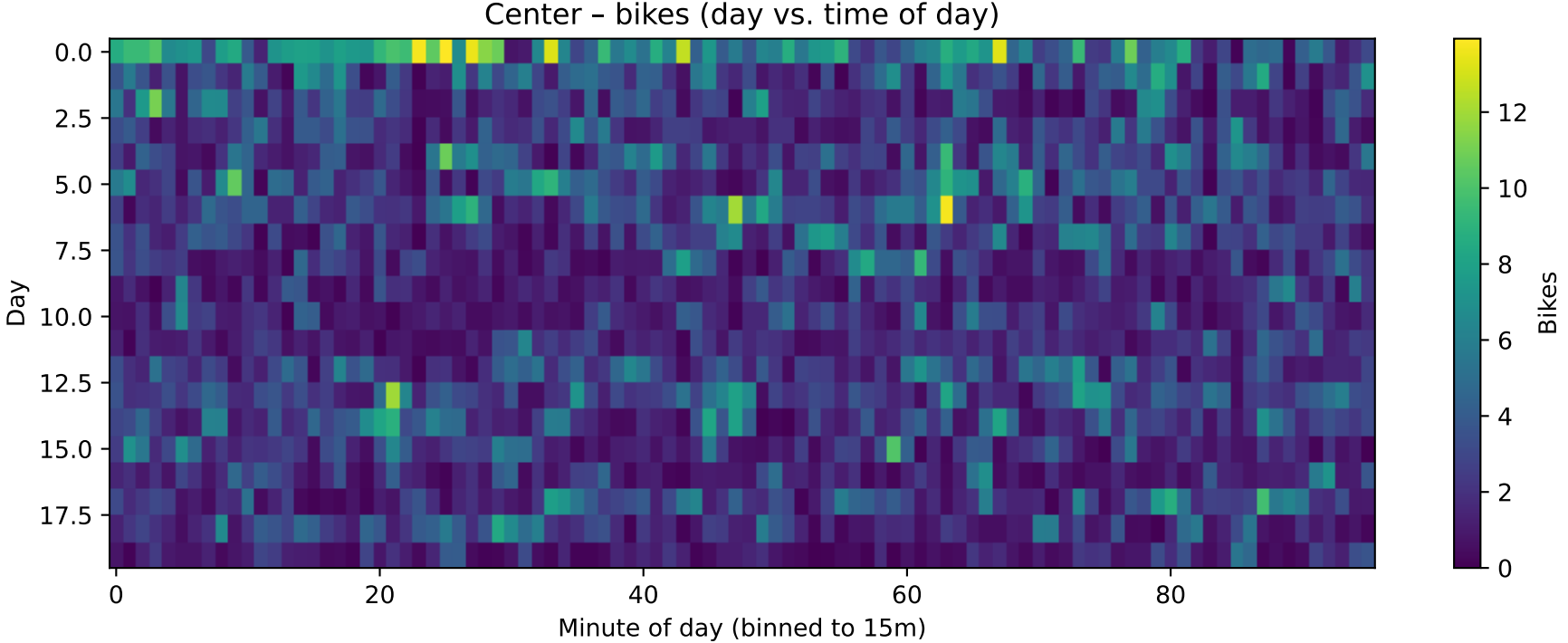
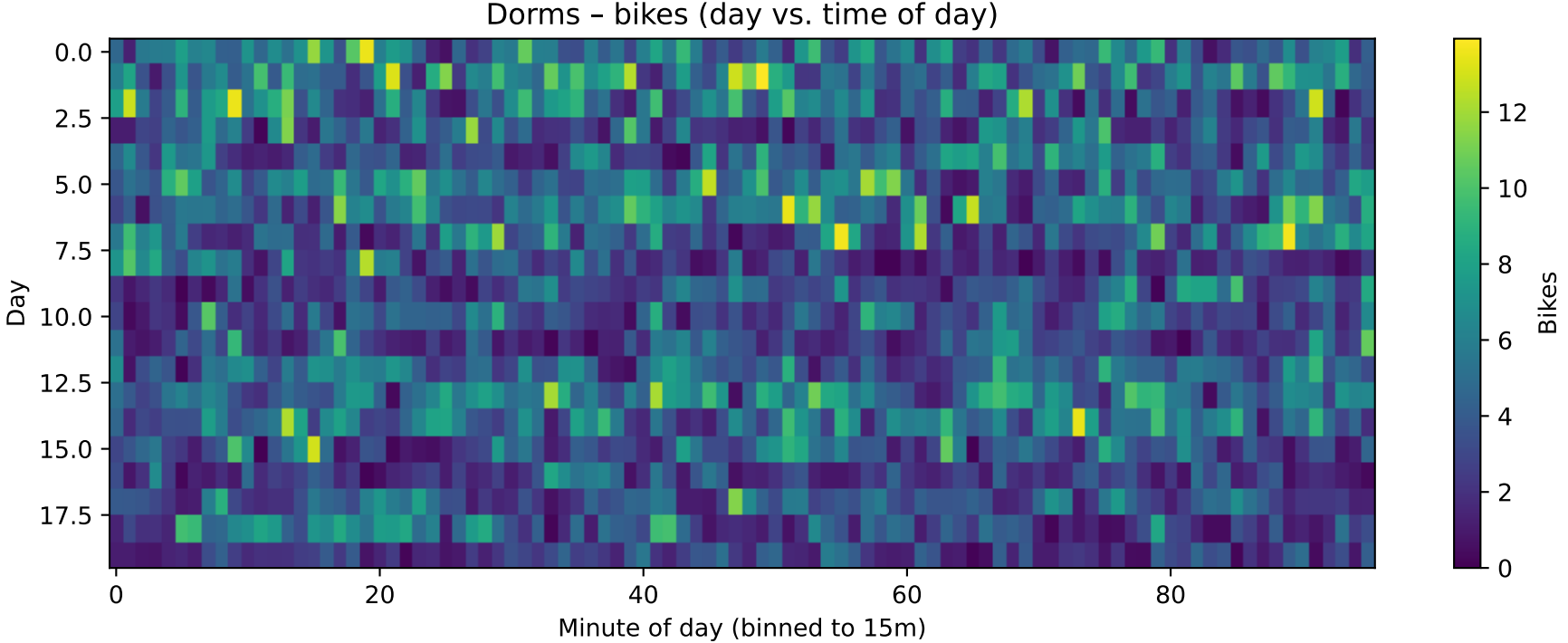
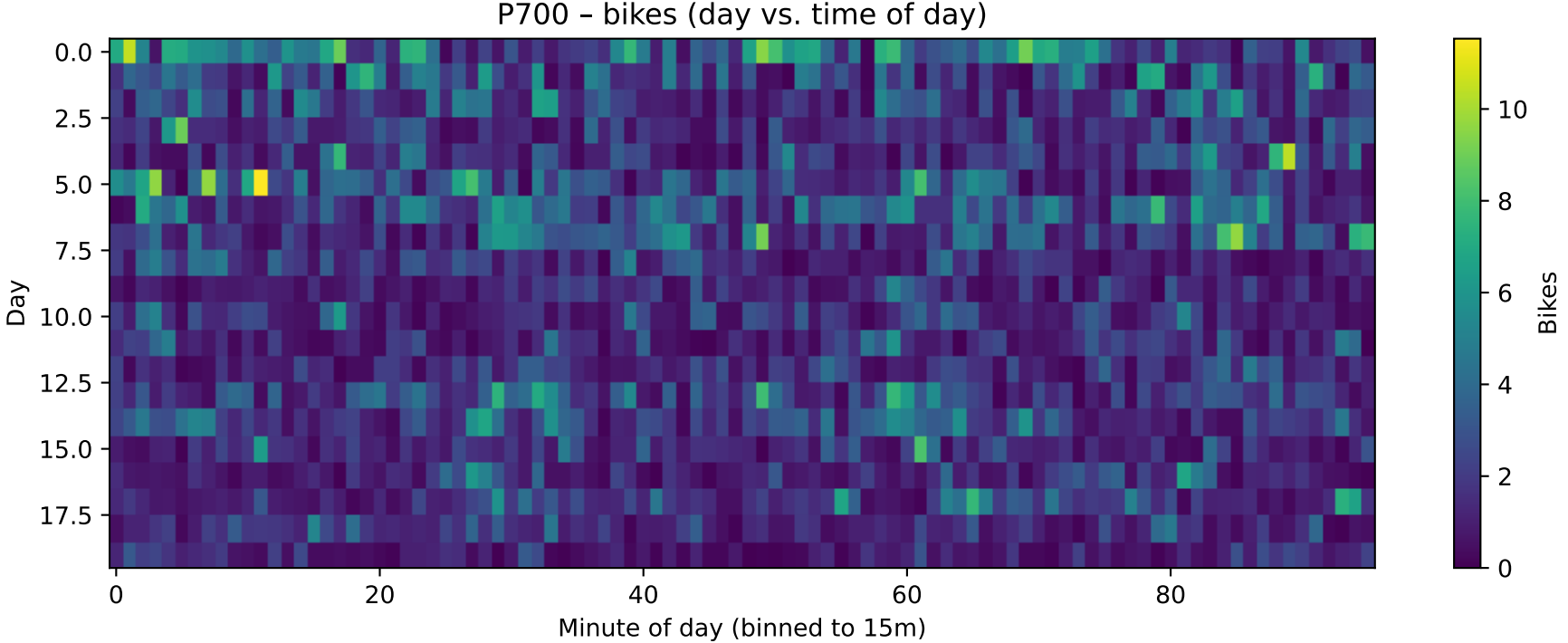
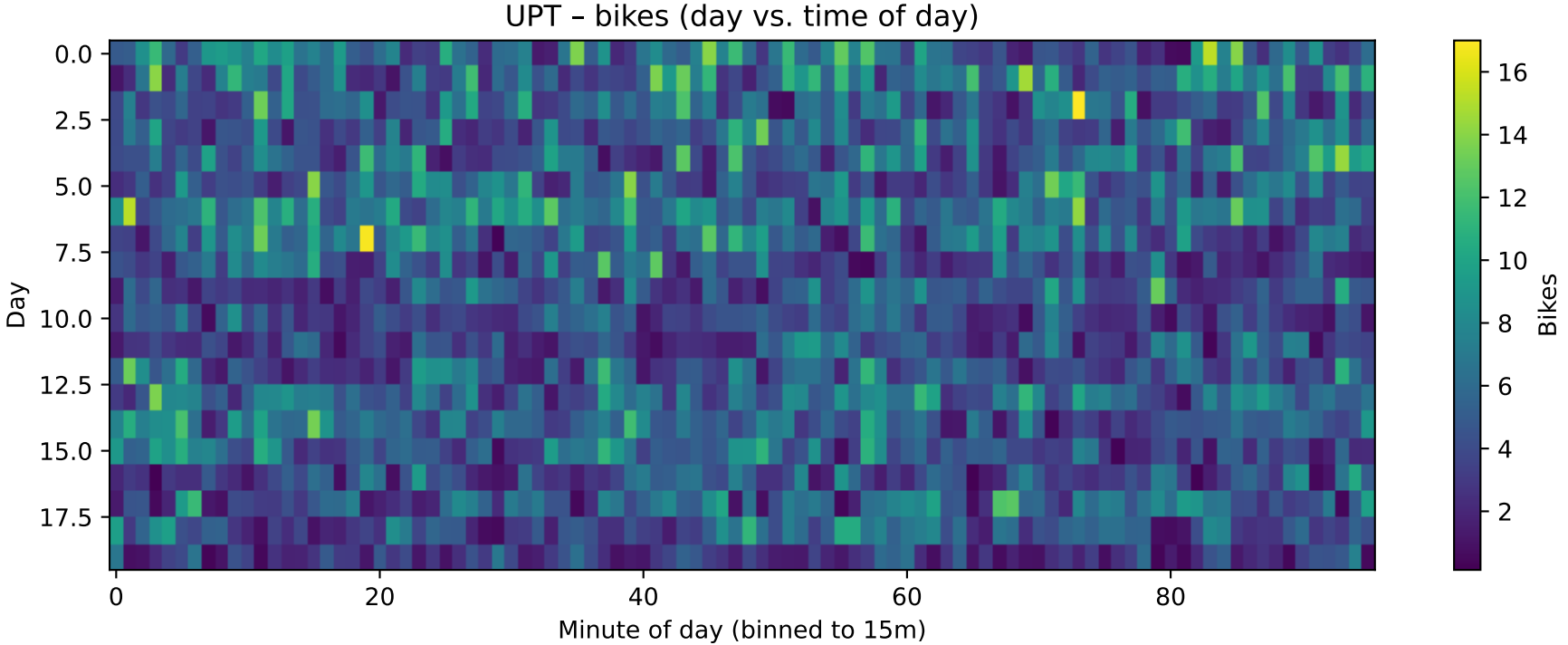
Simulation parameters:
- Duration: 20.0 days (28800 minutes)
- Rebalancing every: 30 minutes (0.020833333333333332 days)
- Breakdown chance: 0.9%
- Repair time: 30 minutes
- Scrapped chance: 0.1%
- Trip time: 2 to 6 minutes
- Probabilities Matrix used: v2

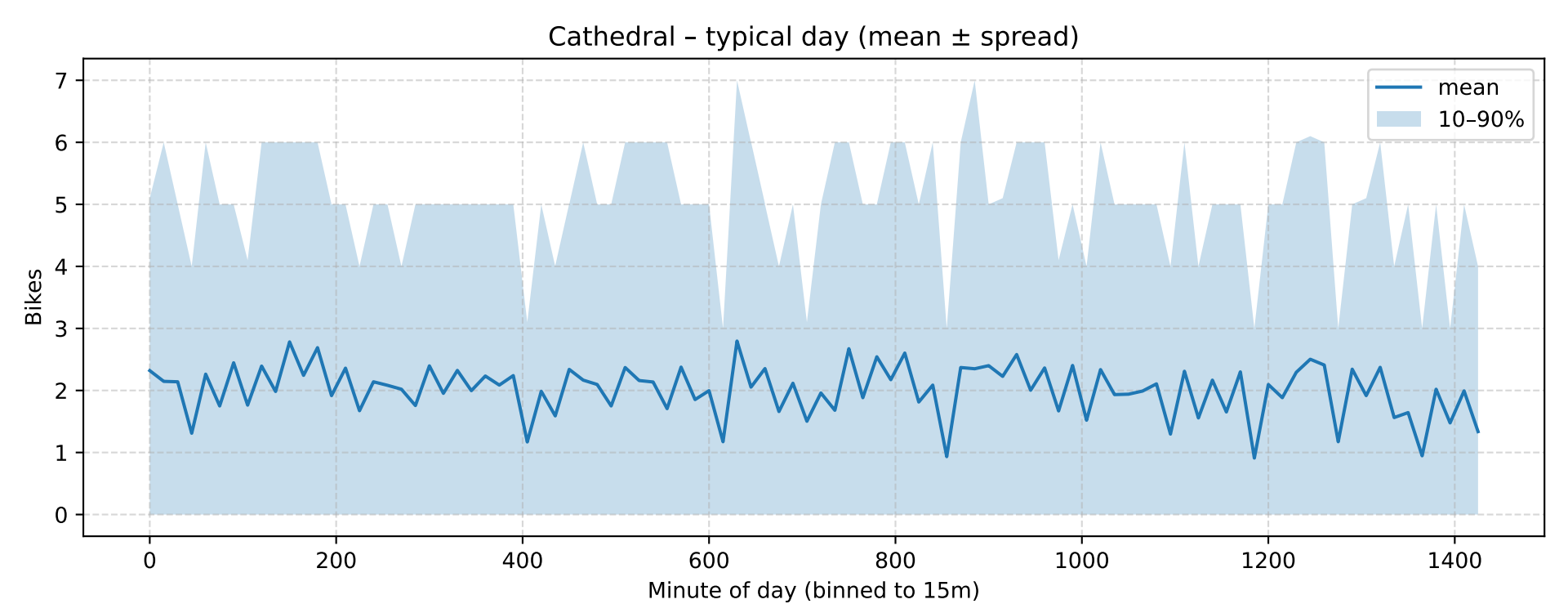

Cathedral – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
0
2
4
6
8

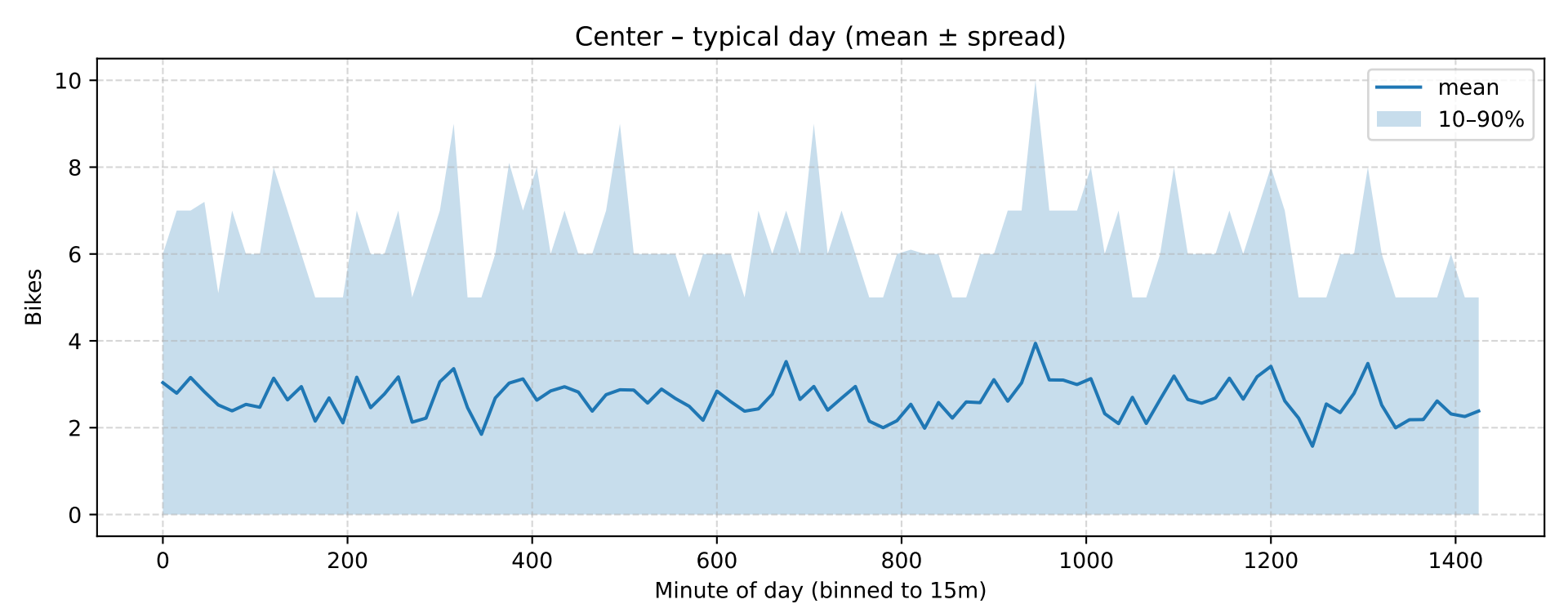

Center – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
0
2
4
6
8
10
12

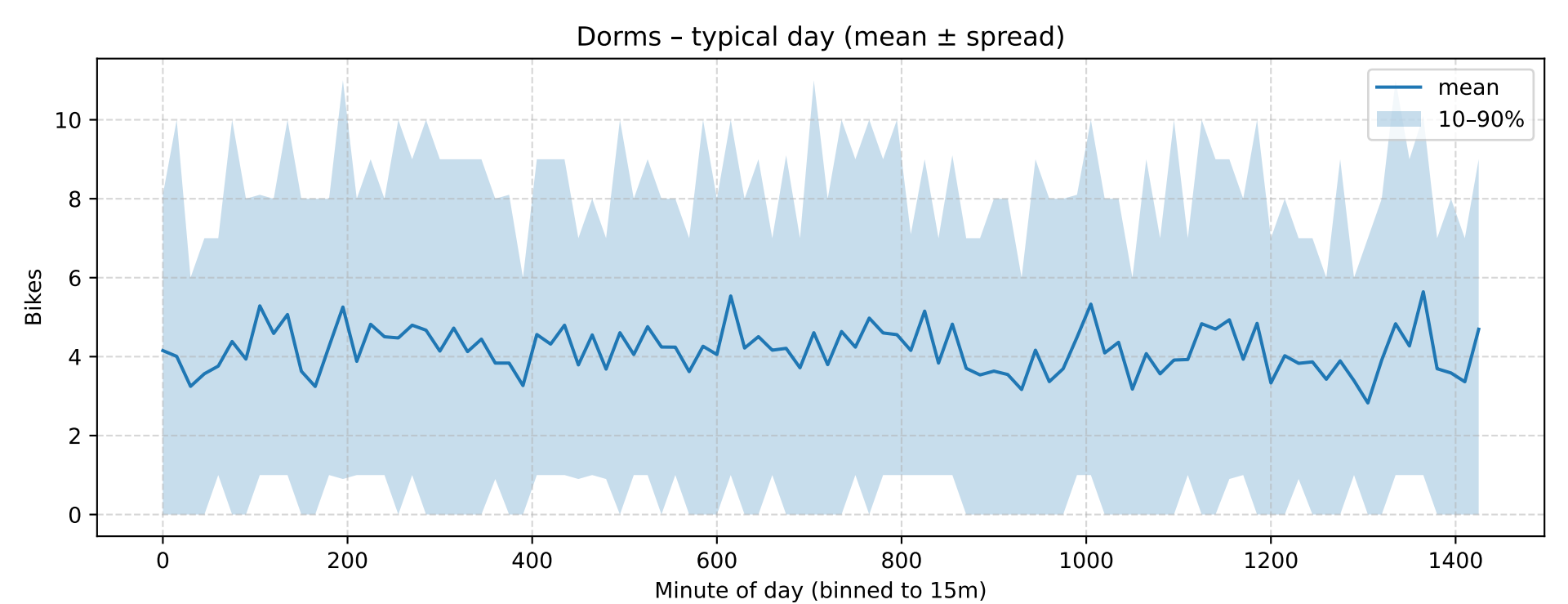

Dorms – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
0
2
4
6
8
10
12

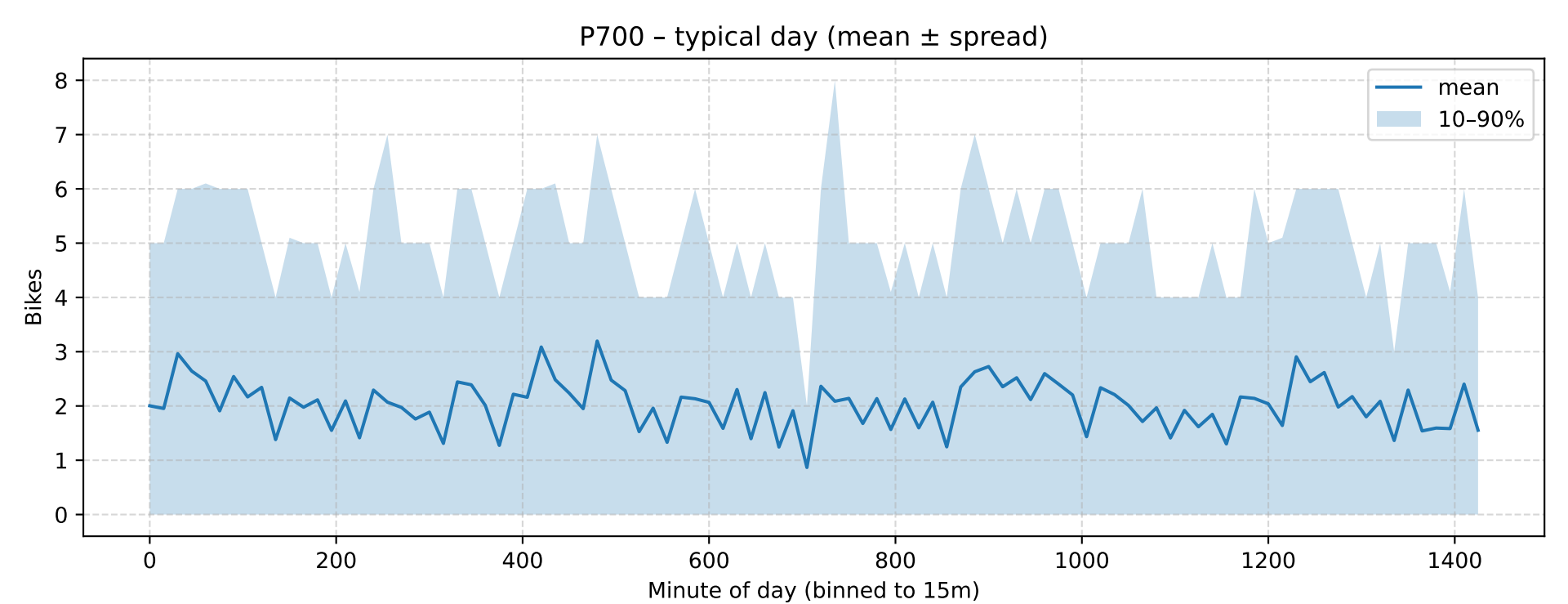

P700 – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
0
2
4
6
8
10

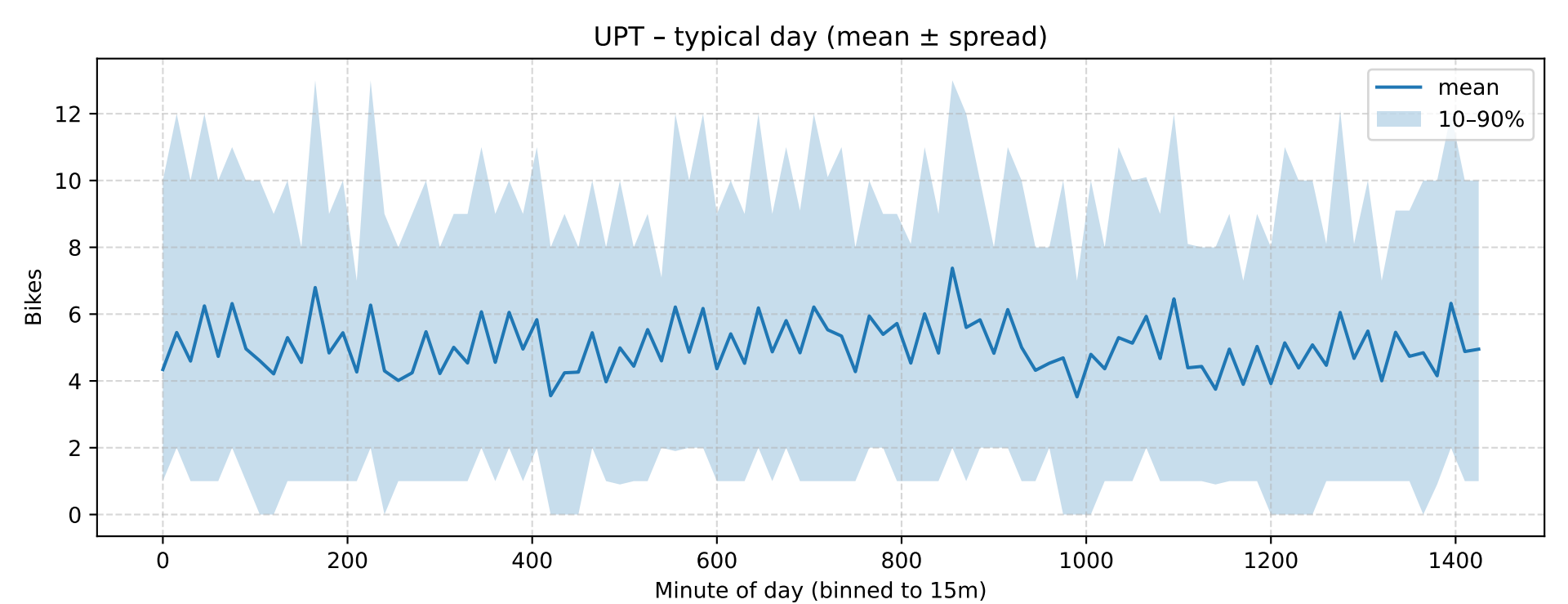

UPT – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
16
14
12
10
8
6
4
2


Cathedral – typical day (mean ± spread)
mean
10–90%
Bikes
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)


Center – typical day (mean ± spread)
mean
10–90%
Bikes
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)

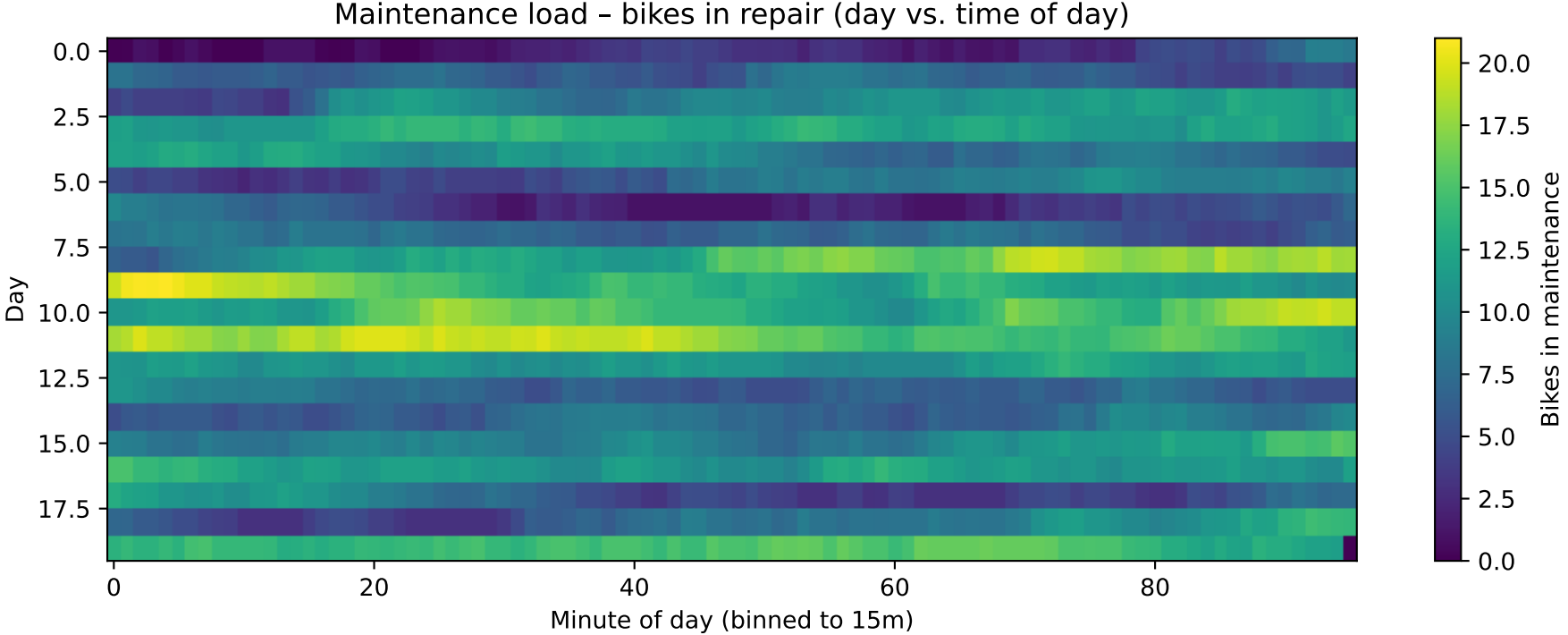

Dorms – typical day (mean ± spread)
mean
10–90%
Bikes
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)

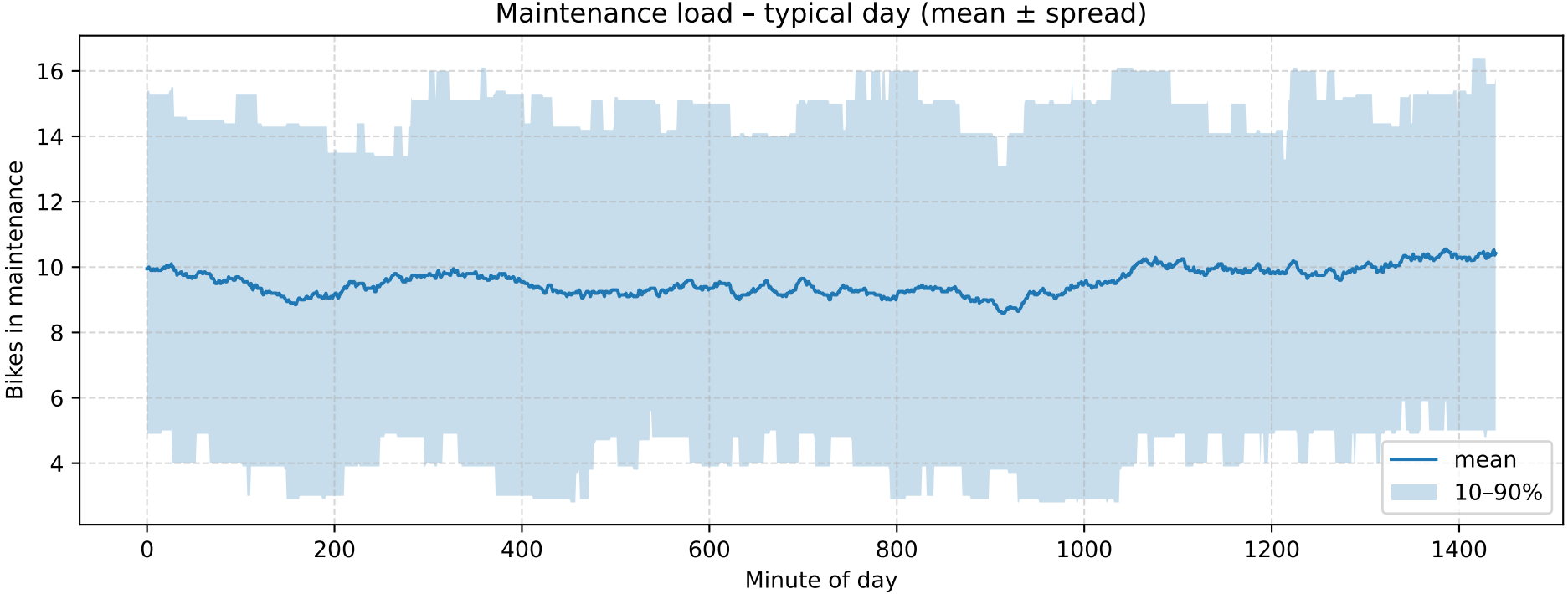

P700 – typical day (mean ± spread)
mean
10–90%
Bikes
0
1
2
3
4
5
6
7
8
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)

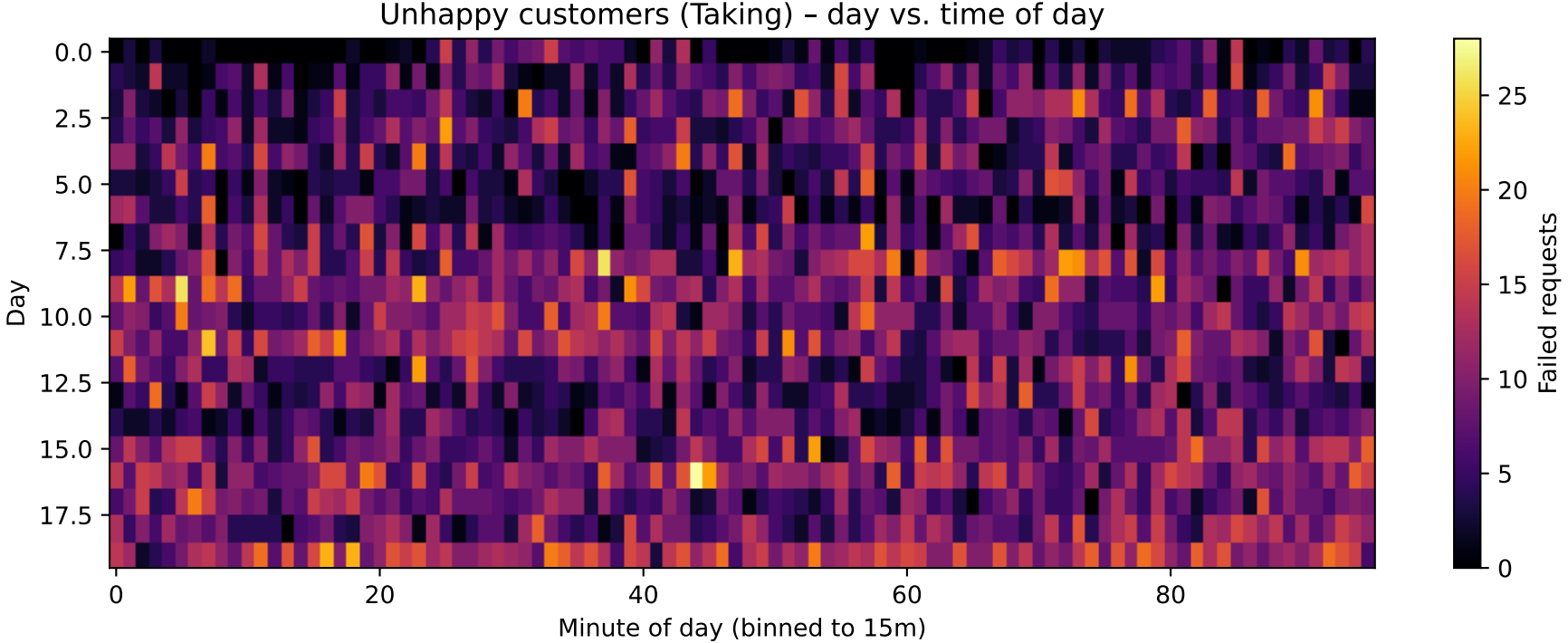

UPT – typical day (mean ± spread)
mean
10–90%
Bikes
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)

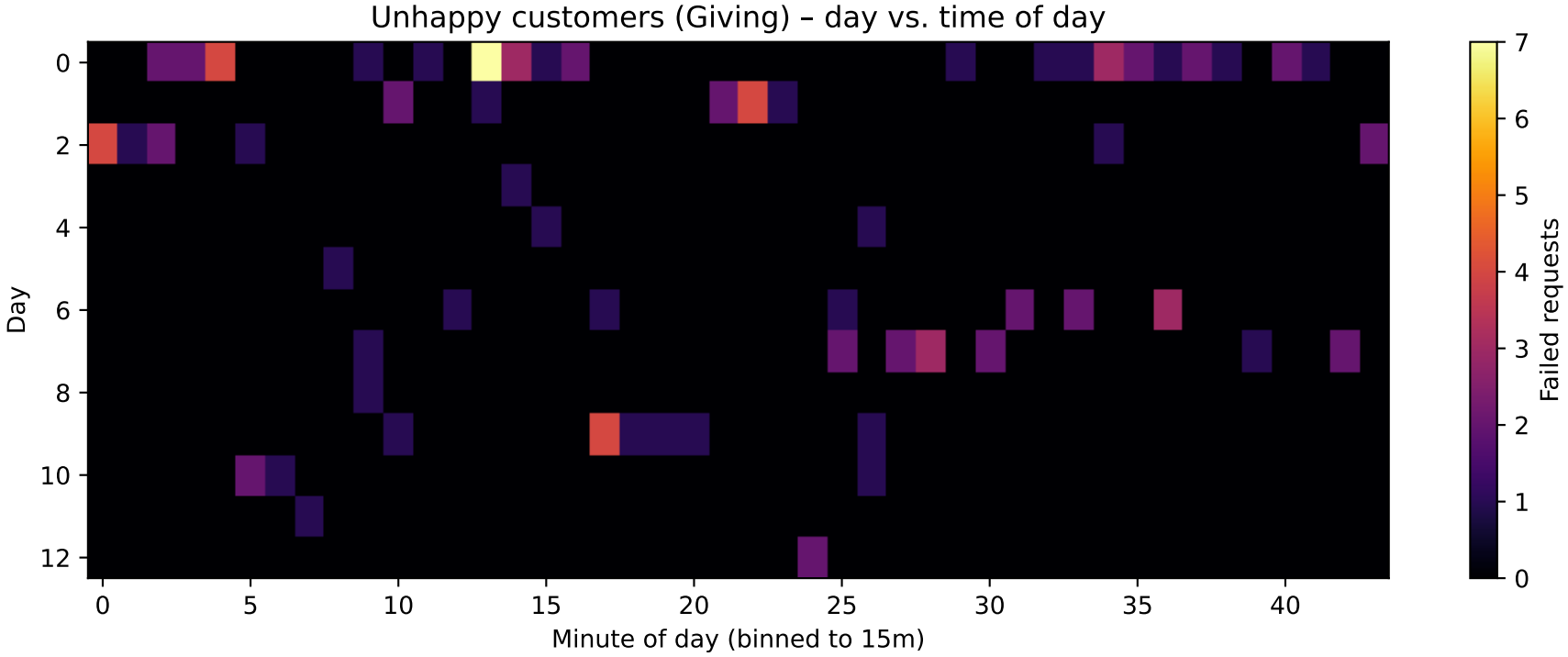
Station KPIs

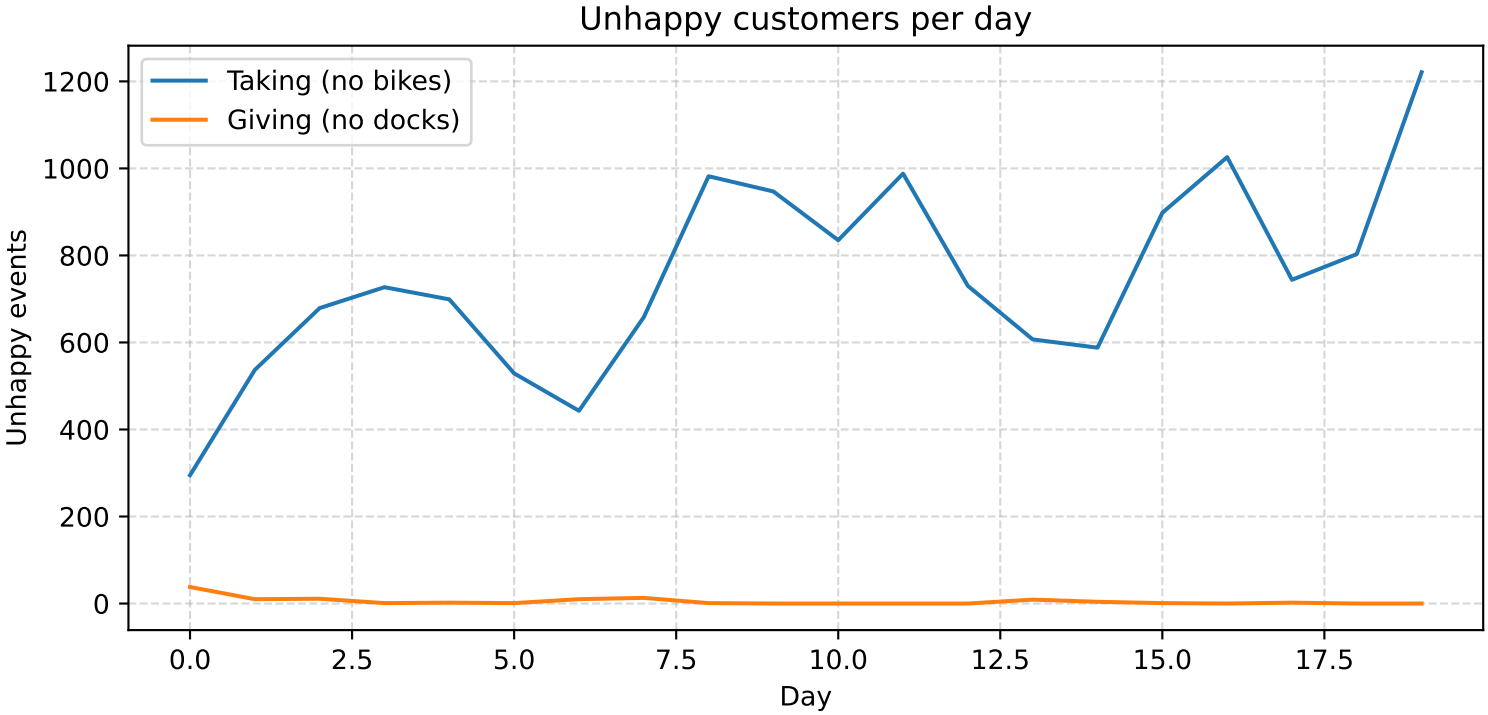
station	avg_bikes	p10_bikes	p90_bikes	stockout_%	overflow_%
Cathedral	2.015	0.0	5.0	33.972	0.236
P700	2.03	0.0	5.0	33.868	0.0
Center	2.658	0.0	6.0	25.497	0.076
Dorms	4.174	0.0	8.0	11.708	0.274
UPT	5.039	1.0	10.0	6.806	0.003

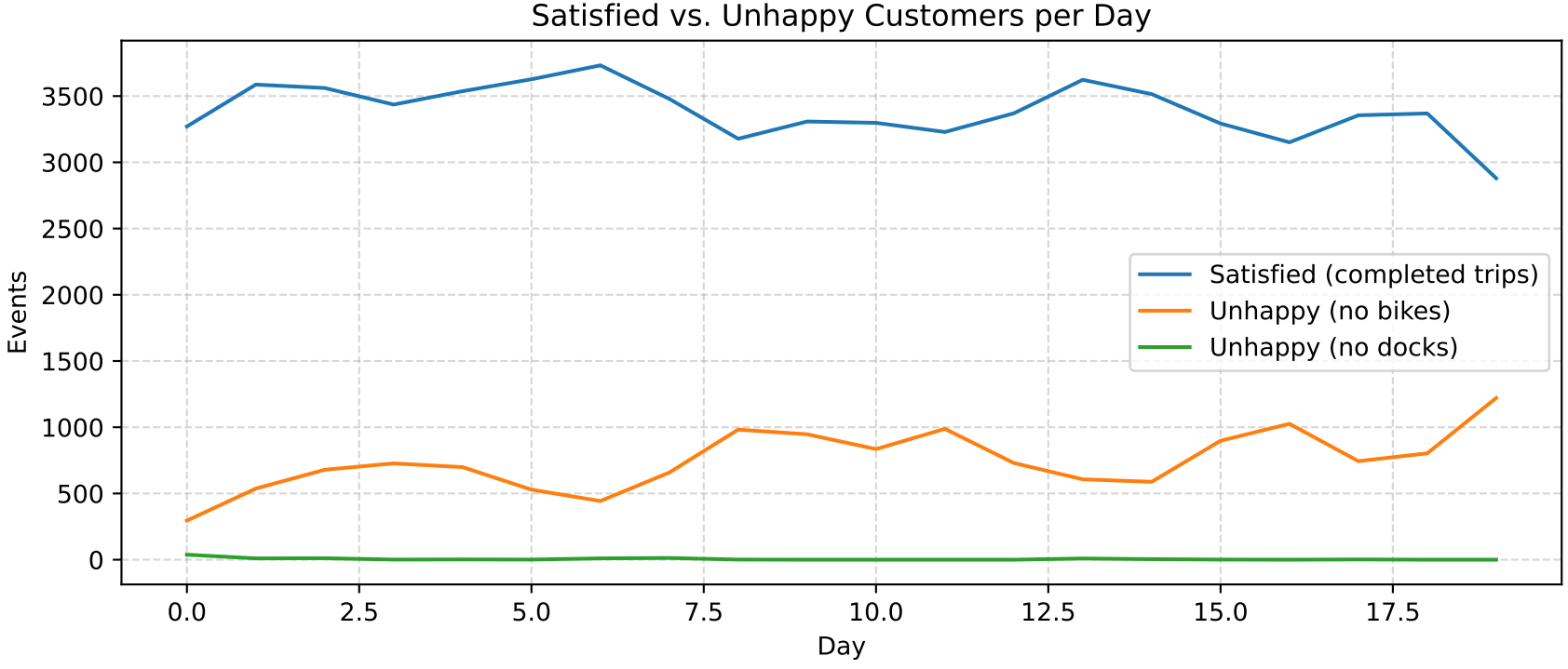

Maintenance load – bikes in repair (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes in maintenance
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0


Maintenance load – typical day (mean ± spread)
Bikes in maintenance
16
14
12
10
8
6
4
0
200
400
600
800
1000
1200
1400
Minute of day
mean
10–90%


Unhappy customers (Taking) – day vs. time of day
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Failed requests
0
5
10
15
20
25


Unhappy customers (Giving) – day vs. time of day
Day
0
2
4
6
8
10
12
0
5
10
15
20
25
30
35
40
Minute of day (binned to 15m)
Failed requests
0
1
2
3
4
5
6
7


Unhappy customers per day
Taking (no bikes)
Giving (no docks)
Unhappy events
1200
1000
800
600
400
200
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Day


Satisfied vs. Unhappy Customers per Day
Events
Day
Satisfied (completed trips)
Unhappy (no bikes)
Unhappy (no docks)
0
500
1000
1500
2000
2500
3000
3500
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

Total satisfied trips: 67799
Total unhappy attempts: 15039
Satisfaction ratio: 81.85%
Total scrapped bikes: 11